

ICT in Education

EDU430

Final Term

Short Notes by M@!k Hafeez

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Topic # 111

Binary Number System

It is a counting system used in computers consisting of only 1's and 0's (zeros).

BinHex

This is a file conversion format that converts binary files to ASCII text files.

Topic # 112

BIOS

This is abbreviation of Basic Input Output System. Part of the computer's operating system that is built into the machine, rather than read from a disk drive at startup.

Bit

It is a unit of measurement that represents one figure or character of data. A bit is the smallest unit of storage in a computer. Since computers actually read 0s and 1s, each is measured as a bit.

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The letter **A** consists of 8 bits which amounts to one byte. Bits are often used to measure the capability of a microprocessor to process data, such as 16-bit or 32-bit.

Topic # 113

Bit Depth

This is regarded as the number of bits used to represent each pixel in an image, determining its color or tonal range.

Bit-map

It is generally used to describe an illustration or font file as being created by a predefined number of pixels. Also see **Object-oriented**.

Topic # 114

BITNET

This is regarded as an academic computer network that provides interactive electronic mail and file transfer services, using a store-and-forward protocol, based on IBM Network Job Entry protocols. BITNET-II encapsulates the BITNET protocol within IP packets and depends on the Internet to route them. There are three main constituents of the network: BITNET in the United States and Mexico, NETNORTH in Canada, and EARN in Europe. There are also Asia Net, in Japan, and connections in South America. See CREN.

Black Point

It is a movable reference point that defines the darkest area in an image, causing all other areas to be adjusted accordingly

Topic # 115

Bootting

This is defined as starting up a computer via the power switch, which loads the system software into memory. Restarting the computer via a keystroke combination is called rebooting or a warm boot.

Bridge

It is a dedicated computer used to connect two different networks. It uses data link layer address (i.e., Ethernet physical addresses) to determine if packets should be passed between the networks.

Best practices in implementing education management information systems (EMIS)

Forty World Bank education projects over the last four years have had EMIS components, but little is known about best practices and lessons learned from such investments. Case studies on EMIS planning and deployment, as well as best practices and lessons learned, would be useful planning tools for donor staff and educational policymakers.

Topic # 116

Broadband System

It is a broadband system is capable of transmitting many different signals at the same time

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without interfering with one another. For local area networks, a broadband system is one that handles multiple channels of local area network signals distributed over Cable Television (CATV) hardware.

Broadcast

This is a packet whose special address results in its being heard by all hosts on a computer network.

Topic # 117

Browser

This is a program that enables you to access information on the Internet through the World Wide Web.

Bps

It may be defined as, Bits Per Second is the unit used for measuring line speed, the number of information units transmitted per second.

Topic # 118

BSD

This is abbreviation of Berkeley Software Distribution. Implementation of the UNIX operating system and its utilities developed and distributed by the University of California at Berkeley. **BSD** is usually preceded by the version number of the distribution, e.g., **4.3 BSD** is version 4.3 of the Berkeley UNIX distribution. Many Internet hosts run BSD software, and it is the ancestor of many commercial UNIX implementations.

Bug

It is a mistake, or unexpected occurrence, in a piece of software or in a piece of hardware.

Topic # 119

Byte

This is regarded as the amount of memory needed to store one character such as a letter or a number. It is equal to 8 bits of digital information. It is the standard measurement unit of a file size.

Cache

This is an area of RAM reserved for data recently read from disk, which allows the processor to quickly retrieve it if it's needed again.

Topic # 120

Caching

It is a process in which frequently accessed data is kept on hand, rather than constantly being from the place where it is stored.

Case-dependent

This is a software differentiation between upper and lower case characters. It is also referred to as case sensitive.

Educational content for dissemination through ICTs

Although it is widely believed to be important, little research has been done about the need for

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gender and culturally inclusive electronic educational materials. Research into how such educational materials can be produced, resulting in best practice guidelines, would be quite useful.

Topic # 121

Define CCD

It is charge coupled device. An integrated, micro-electrical light sensing device built into some image capturing devices.

Define Resampling

It is an increase or reduction in the number of pixels in an image, required to change its resolution without altering its size. See also down-sampling and interpolation.

Topic # 122

CGI

This is a common Gateway Interface. It is a method of running an executable script or program from a Web server. When a client requests a URL pointing to a CGI, the program is run and the results are returned to the client. This enables dynamic web pages and the ability to do database queries and other complex operations across the web

CIE

This is the "Commission Internationale de l'Eclairage". It is an organization that has established a number of widely-used color definitions.

Topic # 123

Circuit Switched

It is a type of network connection which establishes a continuous electrical connection between calling and called users for their exclusive use until the connection is released. Ericsson PBX is a circuit-switched network.

Clickable Image

This may be defined as any image that has instructions embedded in it so that clicking on it initiates some kind of action or result. On a web page, a clickable image is any image that has a URL embedded in it.

Topic # 124

Client/Server Relationship

A client application is one that resides on a user's computer, but sends requests to a remote system to execute a designated procedure using arguments supplied by the user. The computer that initiates the request is the client and the computer responding to the request is the server. Many network services follow a client and server protocol.

Clipboard

This is an area used to temporarily store cut or copied information. The Clipboard can store text, graphics, objects, and other data. The Clipboard contents are erased when new information is placed on the Clipboard or when the computer is shut down.

Topic # 125

Clipping

This is the conversion of all tones lighter than a specified grey level to white, or darker than a specified grey level to black, causing loss of detail. This also applies to individual channels in a color image.

CMS

It is color management system. This ensures color uniformity across input and output devices so that final printed results match originals. The characteristics or profiles of devices are normally established by reference to standard color targets.

Learning strategies fostered by the use of ICTs

Research into the types of teaching and learning practices fostered by ICT use (most especially more 'learner centric' pedagogical strategies) and how such pedagogies affect disadvantaged students would also be quite useful.

Topic # 126

CMYK

It is cyan, magenta, yellow, and black are the base colors used in printing processes. CMY are the primary colorants of the subtractive color model.

Codec

This is also called a compressor, a compression / decompression software component which translates video between its uncompressed form and the compressed form in which it is stored on media (disk, etc.). The two most commonly used video codecs for CD-ROM video are Cinepak and Indeo.

Topic # 127

Colorimeter

This is a light-sensitive device for measuring colors by filtering their red, green, and blue components, as in the human eye. See also spectro-photo-meter.

Color Cast

This is an overall color imbalance in an image, as if viewed through a colored filter.

Topic # 128

COM1, COM2, etc. -

Most serial ports and internal modems on DOS/WIN PCs can be configured to either COM1 or COM2 in order to accommodate the situation where both may exist. The DOS MODE command is used to change the output direction to such serial devices as modems.

Compression

This may be referred as the reduction in size of an image file. See also lossy and non-lossy.

Topic # 129

Computer Search Service

It is Computer Search Service (CSS), a fee-based service, offers access to more than 500

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databases, primarily in the sciences and social sciences, from which the Library can create customized bibliographies.

Configuration

These may be defined as the components that make up a computer system (which model and what peripherals). This is the physical arrangement of those components (what's placed and where). It is the software settings that enable two computer components to talk to each other (as in configuring communications software to work with a modem).

Topic # 130

Contone (CT)

It is an abbreviation for continuous tone. It is a color or grey scale image format capable of illustrating continuously varying tonal ranges, as opposed to line art.

Cookies

It is a file sent to a web browser by a web server that is used to record one's activities.

Emotional, psychological and cultural impacts of ICT on Communities

The impact of ICT use on learners may be most pronounced not on student achievement, but rather on a learner's sense of self and cultural identity.

Topic # 131

Coprocessor

It is a chip designed specifically to handle a particular task, such as math calculations or displaying graphics on-screen. A coprocessor is faster at its specialized function than the main processor is, and it relieves the processor of some work. A coprocessor can reside on the motherboard or be part of an expansion card, as with an accelerator.

Coaxial Cable

It is a type of cable that contains two conductors. The center conductor is surrounded by a layer of insulation, which is then wrapped by a braided-metal conductor and an outer layer of insulation.

Topic # 132

CPU

This is Central Processing Unit; the brains of the computer. The CPU interprets and executes the actual computing tasks.

Crash

It is a problem (often caused by a bug) that causes a program, or the entire operating system, to unexpectedly stop working

Topic # 133

CREN

It is abbreviation of Corporation for Research and Educational Networking. This organization was formed in October 1989, when BITNET and CSNET (Computer + Science Network) were combined under one administrative authority. CSNET is no longer operational, but CREN still runs BITNET. See BITNET.

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Cross Platform

It refers to software (or anything else) that will work on more than one platform (type of computer).

Topic # 134

CSUNET

It is California State University Network. This is a packet-switched network that connects the 22 campuses of the CSU system.

Cursor

It is the representation of the mouse on the screen. It may take many different shapes. Example: I-beam, arrow pointer, and hand.

Topic # 135

Cyberspace

It is a term used to refer to the electronic universe of information available through the Internet.

DAT

This is abbreviation of Digital Audio Tape. It is the most common type of tape backup.

Differential impacts of ICT on education

It is well established that ICTs used in education can have differential impact on boys and girls. However, such research has neglected to explore the possible impact of ICTs on certain sub groups of girls (and boys), and the resulting implications for impact of ICT in education on different types of girls and boys.

Topic # 136

Daughterboard

This is a board that attaches to (rides piggyback on) another board, such as the motherboard or an expansion card. For example, you can often add a daughter card containing additional memory to an accelerator card.

Database

It is a file created by a database manager that contains a collection of information organized into records, each of which contains labeled categories (called fields).

Topic # 137

DCS

This is abbreviation of Desktop Color Separation. It is an image format consisting of four separate CMYK PostScript files at full resolution, together with a fifth EPS master for placement in documents.

DDN

This is abbreviation of Defense Data Network. A global communications network serving the US Department of Defense composed of MILNET, other portions of the Internet, and classified networks which are not part of the Internet. The DDN is used to connect military installations and is managed by the Defense Information Systems Agency (DISA).

Topic # 138

DEC

This is abbreviation of Digital Equipment Corporation.

Decompression

It is the expansion of compressed image files.

Topic # 139

Dedicated line

It is a telephone or data line that is always available. For example, a leased telephone line can be dedicated for computer data communications. This line is not used by other computers or individuals, is available 24 hours a day, and is never disconnected.

Default Route

This is a routing table entry which is used to direct packets addressed to networks not explicitly listed in the routing table.

Topic # 140

Densitometer

It is a measuring instrument that registers the density of transparent or reflective materials. Colors are read as tonal information.

Density

Density is a brightness control to lighten or darken a printout to more closely reflect its screen appearance and to compensate for deficiencies in toner or paper quality.

Use of ICTs to teach illiterate youth

Issues of equity are central to the discussion of the education-related Millennium Development Goals. For a variety of reasons and in a variety of ways, the introduction of ICTs to benefit education can have profound impact on issues related to Education For All. While for many ICTs offer a promise at greater inclusion of previously marginalized groups (whether marginalized by gender, disability, distance, language, culture, race, age or economic status), their use also brings with it very real dangers of increasing the marginalization of such groups inside the education system.

Topic # 141

De-screening

It may be defined as removal of halftone dot patterns during or after scanning printed matter by defocusing the image. This avoids more patterning and color shifts during subsequent halftone reprinting.

Dialog box

It is a window that displays additional options or questions when a command is chosen.

Topic # 142

Dial-up line

It is a communication connection from your computer to a host computer over standard phone

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lines. Unlike a dedicated line, you must dial the host computer in order to establish a connection. Dial-up line is currently the most popular form of Net connection for the home user.

Dichroic Mirror

This is a special type of interference filter, which reflects a specific part of the spectrum, whilst transmitting the rest. It is used in scanners to split a beam of light into RGB components.

Topic # 143

Digital

It may be defined as data or voltages consisting of discrete steps or levels, as opposed to continuously variable analog data.

Digitizers

It is a machine which converts analog data into digital data on a computer (such as a scanner digitizing pictures or text).

Topic # 144

DIP switches

It is Dual Interface Poll switches allow for either an ON or OFF setting with any number of circuits. DIP switches commonly allow you to change the configuration of a circuit board to suit your particular computer.

Direct connection

This is a permanent communication connection between your computer system (either a single CPU or a LAN) and the Internet. This is also called a leased line connection because you are leasing the telephone connection from the phone company. A direct connection is in contrast to a SLIP/PPP or dial-up connection.

Topic # 145

Direct-To-Plate

It is direct exposure of image data onto printing plates, without the intermediate use of film.

Direct-To-Press

This is elimination of intermediate film and printing plates by the direct transfer of image data to printing cylinders in the press.

Successful and relevant strategies for using ICTs

The development of models and case studies on successful strategies for using ICTs to change pedagogical practices would be useful. Especially useful would be case studies and examples of so-called "multi-channel learning" practices, which focus on enriching the environment by engaging the resources that are available to help effect incremental change by coordinating the various ways to connect learners with information, knowledge, and stimulation, and to mediate those interactions. Such practices may be especially relevant in countries seeking to utilize ICTs to help meet education-related MDGs.

The extent to which teacher training is currently being done with ICTs to meet these challenges is not well documented, especially at the in-service level. It is widely believed that interactive radio may be a cost-effective way of providing regular outreach and support to teachers, especially as it relates to (a) teacher subject knowledge and (b) teacher pedagogical practices. In many countries, radio networks exist that serve non-education sectors (for example,

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in agriculture or community development), but that could be used to benefit education. In addition, where interactive radio is being used, many of them are part of pilot projects, and it is unclear to what extent such initiatives can be mainstreamed and sustained by the education system after the pilot project is completed.

Topic # 146

Directory

It is a system that your computer uses to organize files on the basis of specific information.

Disk Defragmenter

This arranges the blocks of information for a file into adjacent blocks on your hard drive, which may significantly improve the file access times.

Topic # 147

DMAX

It is the point of maximum density in an image or an original.

DMIN

It is the point of minimum density in an image or an original.

Topic # 148

DNS

This is called Domain Name System. A general purpose distributed, replicated, data query service. Its principal use is the lookup of host IP addresses based on host names. The host names are also known as domain names. Some important domains are: .COM (commercial), .EDU (educational), .NET (network operations), .GOV (U.S. government), and .MIL (U.S. military). Most countries also have a domain. For example, .US (United States), .UK (United Kingdom), .AU (Australia).

Domain Name Server

It may be defined as, a computer that converts host names, such as rohan.sdsu.edu to its corresponding IP Address, such as 191.130.1.10. An SDSU computer provides this service any time mail is sent or received and permits users to use TELNET and FTP between SDSU and other sites.

Topic # 149

DOS

This is called as Disk Operating System. The operating system used on IBM personal computers and compatible machines.

Dotted Decimal Notation

It is the convention for writing 32-bit IP Addresses as a set of four 8-bit numbers written in base 10 with periods separating them.

Topic # 150

Down-Sampling

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It is the reduction in resolution of an image, necessitating a loss in detail.

Download

It is a process to retrieve a file from another computer using a modem.

Costs/benefits of situating ICTs in schools

Cost-benefit analyses of situating computers outside of classrooms, but rather in school libraries, teacher training institutions and community tele-centres (including those that are school-based), would be quite useful.

Topic # 151

Dpi

It is defined as Dots Per Inch. It is a measure of the resolution of a printer, scanner, or monitor. It refers to the number of dots in a one-inch line. If it will be more dots per inch, then higher the resolution is.

Driver

It is a piece of software that tells the computer how to operate an external device, such as a printer, hard disk, CD-ROM drive, or scanner. For instance, you can't print unless you have a printer driver. Hard disk drivers are invisible files that are loaded into memory when you start the computer, while scanner drivers are usually plug-ins accessed from within a particular application.

Topic # 152

Drum Scanner

Early drum scanners separated scans into CMYK data, recording these directly onto film held on a second rotating drum.

DTP

It is called Desktop Publishing.

Topic # 153

Dump

It is Back up of data.

Duplex (Full, Half)

Full duplex is data is flowing in both directions at the same time. When Remote echo is ON communication is occurring in full duplex. Half Duplex has data moving in only one direction at a time (Local echo is ON).

Topic # 154

DXF

Drawing Interchange Format used for Macintosh graphic files. It is the standard file-exchange format for 3-D and CAD programs.

Dye Sublimation

This is a printing process using small heating elements to evaporate pigments from a carrier film, depositing these smoothly onto a substrate.

Topic # 155

EARN

It is abbreviation of European Academic Research Network.

Echo (Local, Remote)

It may be defined as Local echo ON causes all transmitted data to be sent to the screen of the sending computer. Remote echo ON causes everything that the remote computer (the one you are communicating with) transmits to be duplicated on your computer's screen.

Public private partnerships in cutting 'cut costs' for ICT

Significant work needs to be done related to the costs of ICT in education. All of the claims listed in the related "Knowledge Map" pertaining to costs found in current literature deserve additional scrutiny.

There is little credible data related to the costs of using ICTs to support education in developing countries. Few good, reliable cost studies of ICT in education implementations exist. Those that do exist measure different things, and thus are not easily comparable. While the impact of ICT interventions on student achievement may be difficult to measure, costs explicit and implicit costs of ICT-related educational activities should be measurable.

Topic # 156

EBCDIC

It is abbreviation of Extended Binary-Coded Decimal Interchange Code. It is a standard character-to-number encoding used primarily by IBM computer systems.

E-mail

It is called Electronic Mail. Private messages sent between users on different computers, either over a network or via a modem connection to an on-line service or BBS.

Topic # 157

Encoding

It is the file transfer formatting that enables encrypted, compressed or binary files to be transferred without corruption or loss of data.

Encryption

It is a way of coding information in a file or e-mail message so that if it is intercepted by a third party as it travels over a network it cannot be read.

Topic # 158

EPS

This is abbreviation of Encapsulated PostScript. An EPS file usually has two parts: a PostScript (text) description that tells a PostScript printer how to output the resolution-independent image, and (optionally) a bit-mapped PICT image for on-screen previews. (EPS files without a PICT preview is usually displayed as a gray rectangle.) EPS files generally can't be edited, even by the program that created them (Illustrator files are exceptions).

EPS 5
It is called another term used for DCS.

Topic # 159

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ERIC

It is an index to educational journal articles and documents collected by national clearing houses for educational information. It provides summaries of the articles and documents, and contains citations back to 1966.

Ethernet

This is an IEEE 802.3 standard data link layer which can operate over several different media including fiber optic, coaxial cable and twisted-pair cable. This 10 million-bit-per-second networking scheme is widely used on campus because it can network a wide variety of computers; it is not proprietary; and components are widely available from many commercial sources

Topic # 160

Executable File

It refers to a file that is a program. It is the executable in DOS and Windows usually have an .exe or a .com extension. In UNIX and Macintosh environments, executable files can have any name.

External Viewer

Program used for presenting graphics, audio and movies while browsing World Wide Web pages via a Web client program. Helper applications are another term for these external programs.

Components of ICT in education

If the presence, magnitude and nature of ICT investments in education in developing countries is of interest to donors, standard definitions of ICTs in education must be developed, agreed to, disseminated and utilized, and changes must be made to current coding practices of donor supported education projects. To this end, the formation of a multi-donor task force to study and this issue and agree upon certain common definitions and coding standards could be useful. Such standards and definitions need not (initially) be comprehensive, but should not focus exclusively on hardware indicators. A comprehensive study of a small number of donor-supported education projects to identify all uses of ICTs components could be a useful tool to support the effort to articulate and agree upon common definitions and standards.

Topic # 161

FAQ

This is abbreviation of Frequently Asked Questions. It is a document that covers a topic of general concern to many users. FAQs are a good way for new users to get information on various aspects of the Internet.

FDDI

It is abbreviation of Fiber Distributed Data Interface. It is an emerging standard for network technology based on fiber optics that has been established by the American National Standards Institute (ANSI). FDDI specifies a 100-million bit per second data rate.

Topic # 162

File

It is a collection of information on a disk, usually a document or a program, that's lumped together and called by one name.

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File Permissions

When you place files on a UNIX system you can assign the files various levels of permission, specifying who can access them, and what type of access they can have.

Topic # 163

File Server

It is a computer that shares its resources, such as printers and files, with other computers on the network. An example of this is a Novell NetWare Server which shares its disk space with a workstation that does not have a disk drive of its own.

Film Recorder

It is used in reference to color transparency recording devices, and sometimes also to image setters.

Topic # 164

Filter

It is a piece of software that an application uses for file-format conversion or special effects. PageMaker, for example, has a filter that lets it import Microsoft Word files, while Photoshop has dozens of filters for special effects (such as image blurring). Filters can be part of the main application or external programs called plug-ins.

Finger

It is a program that displays information about someone on the internet.

Topic # 165

Firewall

It is a mechanism that isolates a network from the rest of the Internet, permitting only specific traffic to pass in and out.

Flaming/Flame War

This is reacting to someone's newsgroup posting or e-mail in a hostile manner by publicly chastising the person or bombarding the person with nasty e-mail.

Access and use of ICTs outside school

The relationship between out of school use of ICTs by students, teachers and administrators to in-school use of ICTs in LDCs should be explored in greater depth. Given that confidence in ICT use and ICT-related abilities in school tend to correlate with access to ICTs outside of school, studies examining the differential impact (on achievement, uses and attitudes) of ICT use in schools in rural and urban settings could provide an important contribution to understandings of equity issues in the uses of ICTs in education.

Topic # 166

Flatbed Scanner

It may be defined as any scanning device that incorporates a flat transparent plate, on which original images are placed for scanning. The scanning process is linear rather than rotational.

Floating-point Processor

It is a special chip that handles sophisticated calculations, such as those used in spreadsheets,

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CAD, and scientific programs.

Topic # 167

Folder

It is an object that can hold other objects, such as other folders and files.

Font

It is the software that creates a typeface on a computer screen.

Topic # 168

Format

It is a process to initialize a disk to prepare it for use. The disk is checked for errors and organized so that data can be recorded and retrieved. Formatting a used disk erases any previously stored information.

FPO

It is abbreviation of For Position Only. A low resolution image placed in a document to indicate where the final version is to be positioned.

Topic # 169

Fragmentation

It is a condition where parts of a file are stored in different locations on a disk. When a file is fragmented, the drive's read/write head has to jump from place to place to read the data; if many files are fragmented, it can slow the drive's performance.

Frame-grabbing System

It is a combination of hardware and software, designed to capture individual frames from video clips for further digital manipulation, or consecutive replay on computer platforms.

Topic # 170

FTP

It is abbreviation of File Transfer Protocol. The Internet standard high-level protocol for transferring files from one computer to another across the network.

FTP site

It is a computer which stores files that can be retrieved using FTP. FTP sites which allow anyone to retrieve files (without having an account on that computer) are known as Anonymous FTP sites.

Effective utilization of ICTs for professional development

The need for trained teachers is great throughout most LDCs, and ICTs are increasingly seen as important tools in reaching and engaging teachers on an on-going manner.

Topic # 171

Gamma

It is the curve that describes how the middle tones of your images appear. Often incorrectly referred to as "brightness" and/or "contrast", gamma is a non-linear function. Changing the value

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of the gamma affects middle tones while leaving the white and black of the image alone. It is used to compensate for differences between Macintosh and PC monitors.

Gamma Correction

It is the correction of tonal ranges in an image, normally by the adjustment of tone curves.

Topic # 172

Gamut

It is the limited range of colors provided by a specific input device, output device or pigment set.

Gang Scanning

It is sequential scanning of multiple originals using the same previously defined exposure setting for each.

Topic # 173

Gateway

It is a special-purpose dedicated computer that attaches to two or more disparate networks and converts data packets from one form to another.

GB

It is called Gigabyte. It is a unit of data storage size which represents 10^9 (one billion) characters of information.

Topic # 174

GB

It is called Gigabit. 10^9 bits of information (usually used to express a data transfer rate; as in, 1 Gigabit/second = 1Gbps).

GCG (Grey Component Replacement)

It is a technique for replacing all the neutral tones of an image with an appropriate amount of black.

Topic # 175

GIF

It is abbreviation of Graphic Interchange Format (pronounced jiff). It is a file compression format developed by CompuServe for transferring graphic files to and from on-line services.

Define Webmaster

It may be defined as a person or group of people who maintain and administer a web server. Webmaster also refers to a standard Email address at most web hosts where comments and questions can be sent.

Impact of Computers on learning

A comparative study of the benefits and costs of placing computers in computer laboratories, individual classrooms, teacher offices, libraries, and/or community centers would be quite useful.

Topic # 176

Gopher

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It is a consistent user interface and gateway into many on-line white pages and other address databases.

Graphical User Interface (GUI)

It is the graphical visual representation of the working environment that presents the elements of your computer as objects on a desktop.

Topic # 177

Grey Balance

It is the balance between CMY colorants required to produce neutral greys without a color cast.

Grey Levels

It is called discrete tonal steps in a continuous tone image, inherent to digital data. Most CT images will contain 256 grey levels per color

Topic # 178

Grey Scale

It is a continuous tone image comprising black, white, and grey data only.

Hacker

It is a slang term which is used for a technically sophisticated computer user who enjoys exploring computer systems and programs, sometimes to the point of obsession.

Topic # 179

Halftone

It is a simulation of continuous tones by the use of black or overlapping process color dots of varying size or position.

Halftoning Factor

It is a multiplication factor (between 1 and 2) applied to output screen ruling to calculate scanning resolution for optimum output quality. This is also known as the halftoning factor.

Topic # 180

Halo

It is a light line around object edges in an image, produced by the USM (sharpening) technique.

Handle

It is unique character string identifier assigned to each entry in the NIC WHOIS database.

Use of ICTs in Classroom

The use of television and LCD projectors are in wide use in many LDCs as information presentations.

Topic # 181

Handshaking

It is the process computers and modems go through in order to establish a connection and agree on the speed and protocols for data transmission.

Hayes Compatible

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It is a modem is considered Hayes Compatible if it supports the Hayes command set (a language used to communicate with and control a modem)

Topic # 182

Header

It is the portion of a packet, preceding the actual data, containing source and destination addresses, error checking and other fields. A header is also the part of an electronic mail message that precedes the body of a message and contains, among other things, the message originator, date and time.

High Key

It is a light image that is intentionally lacking in shadow detail.

Topic # 183

Highlight

It is the lightest tones in an image. It is a spectral highlight is a bright, reflected light source.

Histogram

This is a chart displaying the tonal ranges present in an image as a series of vertical bars.

Topic # 184

Home Page

This is the document that is displayed when you first open a web client program. Also, commonly used to refer to the first document you come to in a collection of documents on a Web site.

Host

It is the main computer system to which users are connected.

Topic # 185

Hostname

It is a name which officially identifies each computer attached to the Internet.

Hotlist

This is a file containing lists of Internet Web pages, enabling you to easily access your favorite Web pages.

Emerging Practices of using ICT in Education

Handheld devices have one great advantage over many other types of ICTs – they are mobile – and anecdotal evidence suggests that this mobility could provide help meet certain needs of schools and educational systems.

Topic # 186

HP

It is abbreviation of Hewlett Packard.

HTML

It is called Hyper Text Markup Language. A system for tagging various parts of a Web

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document that tells the Web client programs how to display the document's text, links, graphics and attached media.

Topic # 187

Hue

This is the color of an object perceived by the eye due to the fact that a single or pair of RGB primary colors predominates.

Hypermedia

This describes hypertext in which various types of data can be stored - sound, images, video and so on - as regular text.

Topic # 188

Hypertext

It is a text-linking strategy that lets you jump between related information in a document by clicking on a button or highlighted word. On-line help systems often use hypertext links, as do some programs designed for the electronic distribution of documents.

Define Scripts

This is a type of program that consists of a set of instructions for another application or utility to use.

Topic # 189

I-beam

This is the blinking vertical line that shows the point at which text or graphics will be inserted.

IBM

It is the abbreviation of International Business Machines Corporation.

Topic # 190

Icon

This is a graphic symbol, usually representing a file, folder, disk or tool.

Image Map

It is a graphic which is divided into regions or "hotspots". When a particular region is clicked, it calls up a web page that has been associated with that particular region.

Existing models of ICTs

The community telecentres movement has grown enormously in the past decade. However, sustainable models for such centres have not emerged. Increasingly, multi-purpose centres based in schools are seen as having a greater likelihood of achieving sustainability, but little data supports such beliefs.

Topic # 191

Image setter

It is a device used to record digital data (images and text) onto monochrome film or offset litho printing plates by means of a single or multiple intermittent light beams. Color separated data is

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recorded as a series of slightly overlapping spots to produce either solid areas of line-art or halftone dots for printing continuous tones.

Import

It is the process to bring data into a document from another document, often generated by a different application.

Topic # 192

Inactive Window

It is a window that is open but is not the top window.

Info Path

It is also called as Internet gopher for the University of California at San Diego.

Topic # 193

Information Technology

It includes matters concerned with the furtherance of computer science and technology, design, development, installation and implementation of information systems and applications.

Initializing (formatting)

This is setting up a disk (any kind) to receive information. When a disk is initialized (formatted), its magnetic media is divided into tracks and sectors, and structure files that your computer uses to keep track of data are created.

Topic # 194

Inline images

These are graphics that are contained within a document's textual information. In a Web document, these graphics can either be loaded automatically when the page is accessed or loaded manually by clicking on the image's icon.

Installer

It is a utility that copies system software or an application from floppy disks or a CD-ROM to your hard disk. An Installer may also decompress the new files, remove obsolete files, place extensions and control panels in their proper folders, and/or create new folders.

Topic # 195

Interface

It is the way a computer interacts with a user or a peripheral.

Internet

The Internet (note the capital I) is the largest internet in the world. It is a three level hierarchy composed of backbone networks (e.g., NSFNET, MILNET), mid-level networks, and stub networks. The Internet is a multiprotocol internet.

Use of Different Network to enhance education

Potential target countries for this study include those with established experience and expertise in using radio for developmental purposes outside the formal education sector.

Topic # 196

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Interpolation

In the image manipulation context, this is the increase of image resolution by the addition of new pixels throughout the image, the colors of which are based on neighboring pixels.

Interrupt

It is a brief interruption of the computer's activity so that an urgent task can be performed.

Topic # 197

IP

Internet Protocol is the standard that allows dissimilar hosts to connect to each other through the Internet. This protocol defines the IP datagram as the basic unit of information sent over the Internet. The IP datagram consists of an IP header followed by a message.

IP Address

Network addresses are usually of two types: (1) the physical or hardware address of a network interface card; for Ethernet this 48-bit address might be 0260.8C00.7666. The hardware address is used to forward packets within a physical network. (2) The logical or IP Address is used to facilitate moving data between physical networks and is made up of a network number, a sub network number, and a host number. All Internet addresses at SDSU have a network number of 130.191, a subnet number in the range of 1-254, and a host number in the range of 1-254

Topic # 198

P Datagram

It is the basic unit of information passed across the Internet. An IP Datagram is to the Internet as a hardware packet is to a physical network. It contains a source and destination address along with data. Large messages are broken down into a sequence of IP Datagrams.

IRC

This is abbreviation of Internet Relay Chat. It is a program that allows you to carry on "live" conversations with people all over the world by typing messages back and forth across the Internet.

Topic # 199

ISAAC

This is abbreviation of Information System for Advanced Academic Computing. It serves as a clearinghouse for information about the use of IBM-compatible hardware and software as aids to instruction and research in higher education. Membership is free to all students, faculty, and staff at these institutions.

ISO

It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

Provision of equipment for ICTs through Public Private partnership

Due to the large costs involved in equipping schools with and maintaining ICTs, innovative public-private-community partnerships are being explored to help reduce costs.

Topic # 200

Traditionally, students attempt programs, after being lectured in theory and are unable to get them to work. We should initially allow them to understand the nature and solutions to basic programming mistakes; syntax or designed based errors. In the slide in front of you it says: “This is one of the skills that could not be developed by students emphasizing program compression and debugging strategies in program courses. Give them examples of a program that is not running according to expectations. If you have programmed you would realize how complex and frustrating a task it is; problems can occur due to multiple reasons. The importance of mistakes; an entire course can be dedicated to this. In our conventional educational system mistakes are considered bad: a student’s marks and grades are adversely affected. In the book, mentioned earlier, “the power of mistakes and struggles,” the basic deals with this subject. When a student commits an intelligent mistake his knowledge increases.

Topic # 207

God is my Only Helper

In the earlier modules we were looking at a case study of a university where I am the dean. We are particularly analyzing the high dropout rate and its reasons; in context of theories of learning, technology and pedagogy. As you know the monsoon season has begun and the weather milder. While lecturing at another place and trying to convey the necessity of passion for teaching and learning. Some topical verses penned by me which related to the weather and passion:

Khuda he hai wahid madadgar mera
Kay dekhay hai mai nay uske jazbo kay toofan
Kay dekhi hain mai nay uske zulfain pereshan
Kay dekhi hain mai nay uske chahatt kay saman
Vo jazbo kay toofan, vo zulfain pereshan, vo chaat kay saman
Vo jazbay gin ki hararat say jal kar uthi hai sawan ki uudi ghattain
Vo zulfain gin say samhi hoi hai shumali samandur ki thandi hawai
Vo chaat jis ki baho mai aakar, uthti nahi hai uppar nighain
God is my only helper
That I have seen her emotions in a storm
That I have seen her troubled tresses
That I have seen her state of love
Her emotions in a storm, her troubled tresses, her state of love
On seeing these, envious purple storms clouds arise
The cold northern winds fearful of her tresses

Remember me in your prayers